Songela W. Chen

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EDUCATION

University of California, Berkeley 2022–
PhD Candidate, Physical Chemistry
Advisor: David T. Limmer

Massachusetts Institute of Technology 2015–2019
SB Chemistry, minor in Chinese
Advisor: Adam P. Willard
Thesis: Modeling ion mobility in solid-state polymer electrolytes

EXPERIENCE

Limmer Group, UC Berkeley — *Graduate Student Researcher* 2022–
Minimize heat dissipation in logical circuits using stochastic thermodynamics and optimal control techniques.

D. E. Shaw Research — *Scientific Associate* 2019–2022
Improve methods to predict protein-ligand binding free energy for computational drug discovery applications.

Willard Group, MIT — *Undergraduate Research Assistant* 2017–2019
Model ion mobility in solid-state polymer electrolytes using coarse-grained molecular dynamics simulations.

D. E. Shaw Research —*Intern* Summer 2018
Develop enhanced sampling methods to compute binding free energies of protein-protein complexes.

Drennan Lab, MIT — *Undergraduate Research Assistant* January–September 2017
Characterize glycyl radical enzymes prominent in the human gut microbiome using X-ray crystallography.

D. E. Shaw Research — *Early College Intern* Summer 2016
Optimize Hamiltonian tempering schemes for molecular dynamics simulations of protein-ligand systems.

Hu Lab, University of Pittsburgh — *Volunteer* 2011–2014
Investigate effect of omega-3 polyunsaturated fatty acids on microglial responses to myelin pathology in murine cell cultures.

PUBLICATIONS

5. **Chen SW**, Limmer DT. Optimal control of bit erasure in stochastic random access memory, arXiv:2601.14387 [cond-mat.stat-mech].
4. Helms P*, **Chen SW***, Limmer DT. Stochastic thermodynamic bounds on logical circuit operation. *Phys Rev E* 111, 034110 (2025).
3. Piana S, Robustelli P, Tan D, **Chen S**, Shaw DE. Development of a Force Field for the Simulation of Single-Chain Proteins and Protein-Protein Complexes. *J Chem Theory Comput* 16, 2494–2507 (2020).
2. **Chen S**, Zhang H, Pu H, Wang G, Li W, Leak RK, Chen J, Liou AK, Hu X. n-3 PUFA supplementation benefits microglial responses to myelin pathology. *Sci Rep* 4, 7458 (2014).
1. Hu X, Li P, Guo Y, Wang H, Leak RK, **Chen S**, Gao Y, Chen J. Microglia/Macrophage Polarization Dynamics Reveal Novel Mechanism of Injury Expansion After Focal Cerebral Ischemia. *Stroke* 43, 3063 (2012).

* denotes equal contribution.

PRESENTATIONS

APS Global Physics Summit, Denver, CO March 2026
†Berkeley Statistical Mechanics Meeting, Berkeley, CA January 2026
*BITS: Physics of Agency, Kirkwood, CA September 2025
Stochastic Thermodynamics and Theoretical Computer Science II, Santa Fe, NM June 2025
*Complexity Sciences Center, Davis, CA May 2025
Workshop on Stochastic Thermodynamics VI, Virtual May 2025

APS Global Physics Summit, Anaheim, CA — March 2025
†Berkeley Statistical Mechanics Meeting, Berkeley, CA — January 2025
APS March Meeting, Minneapolis, MN — March 2024
†Berkeley Statistical Mechanics Meeting, Berkeley, CA — January 2024

* denotes invited presentation; † denotes poster.

TEACHING

University of California, Berkeley

Graduate Student Instructor
CHEM 220A: Thermodynamics and Statistical Mechanics — Fall 2024
CHEM 120B: Physical Chemistry — Fall 2023
CHEM 1A: General Chemistry Laboratory — Fall 2022

SERVICE

Women in Stochastic Thermodynamics — *Co-Organizer, Web Director* 2025–
Plan and coordinate quarterly meetings for women and non-binary researchers in stochastic thermodynamics.

Mathematics Boot Camp for Physical Chemistry — *Organizer, Instructor, Discussion Leader* 2023–2025
Week-long workshop for incoming first-year physical chemistry students. Plan lesson and discussion content, hire and train other instructors and discussion leaders, write funding proposals, advertise to and gather interest from incoming students.

Women of DESRES — *Coordinator* 2019–2022
Organize monthly events for intra-company women's affinity group, including social lunches, book club, and outreach.

Northeast Regional Middle School Science Bowl — *Assistant Director and Co-Founder* 2015–2019
Organize a daylong quiz bowl event for middle school teams from five states. Fundraise, train volunteers, and maintain website. This event was the first Science Bowl competition in the nation run entirely by students.

MIT ClubChem — *President, Community Outreach Coordinator, Publicity Chair, Web Director* 2015–2018
Manage all aspects of the undergraduate association for chemistry students, including chemistry outreach events at K-8 schools, club presentation at USA Science and Engineering Fair in Washington, DC, and intra-department activities.

MISCELLANEOUS

Bay Area Scientists Inspiring Students (BASIS) — *Volunteer* 2025–
Teach interactive one-hour workshop on simple circuits to 4th and 5th grade classrooms.

Berkeley Science Review — *Designer* 2025
Design illustrations for popular science magazine covering UC Berkeley research.

SKILLS

Computational	Monte Carlo simulation, molecular dynamics, Python, JAX, Bash, LaTeX, Git, C++, Mathematica, HTML, CSS, Adobe Illustrator, Adobe InDesign
Laboratory	X-ray crystallography, protein purification, SDS-PAGE, ELISA, cell culture
Natural Languages	English, Mandarin Chinese (fluent), Shanghainese (fluent), Spanish (intermediate)

AWARDS

Kavli Energy NanoScience Institute Graduate Student Fellowship — 2026
National Science Foundation Graduate Research Fellowship Program Honorable Mention — 2022, 2024
Massachusetts Institute of Technology Freshman Chemistry Achievement Award — 2016

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